Report on emerging and future IT skills requirements

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EXECUTIVE SUMMARY

- Data was collected on:
  - Current and future use of IT languages, tools, databases and platforms
  - Job postings for IT professionals
  - Demographics and education of IT professionals
- This data was analyzed to identify emerging and future skills requirements our company must meet to remain a key player in the IT consulting global scenario.
- From this analysis, insights and key trends were obtained, indicating the need of actions on:
  - Workforce retraining
  - Workforce recruitment
  - Work environment assessing and possible changes

INTRODUCTION

- The purpose of this report is to present key insights and trends on “in-demand” skills in the Information Technology global scenario.
- This report is targeted to our company’s Human Resources Division.
- The information herein contained will be essential for HR to define action items to keep our workforce up-to-date with emerging and future IT skills requirements.
- Questions this report intends to answer on use of technology:
  - Which programming languages are more used now?
  - Which programming languages are more likely to be used in the future?
  - Which database technologies are more used now?
  - Which database technologies are more likely to be used in the future?
  - Which web frameworks are more used now?
  - Which web frameworks are more likely to be used in the future?
  - Which cloud platforms are more used now?
  - Which cloud platforms are more likely to be used in the future?

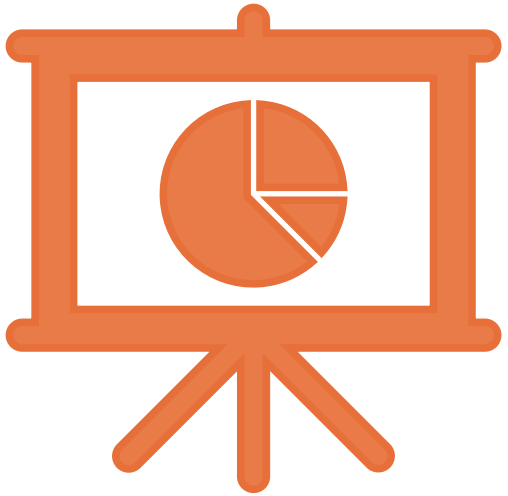
INTRODUCTION (continued)

- Questions this report intends to answer on Job Postings:
  - What is the distribution of job postings by technology?
  - What is the distribution of job postings by location?
- Questions this report intends to answer on compensation:
  - What is the annual compensation for the most popular languages?
  - What is the annual compensation by developer's age ranges?
  - What is the annual compensation by experience levels?
- Questions this report intends to answer on job satisfaction:
  - What is the job satisfaction by experience level?
- Additionally, the charts present in this report can show other trends and insights.

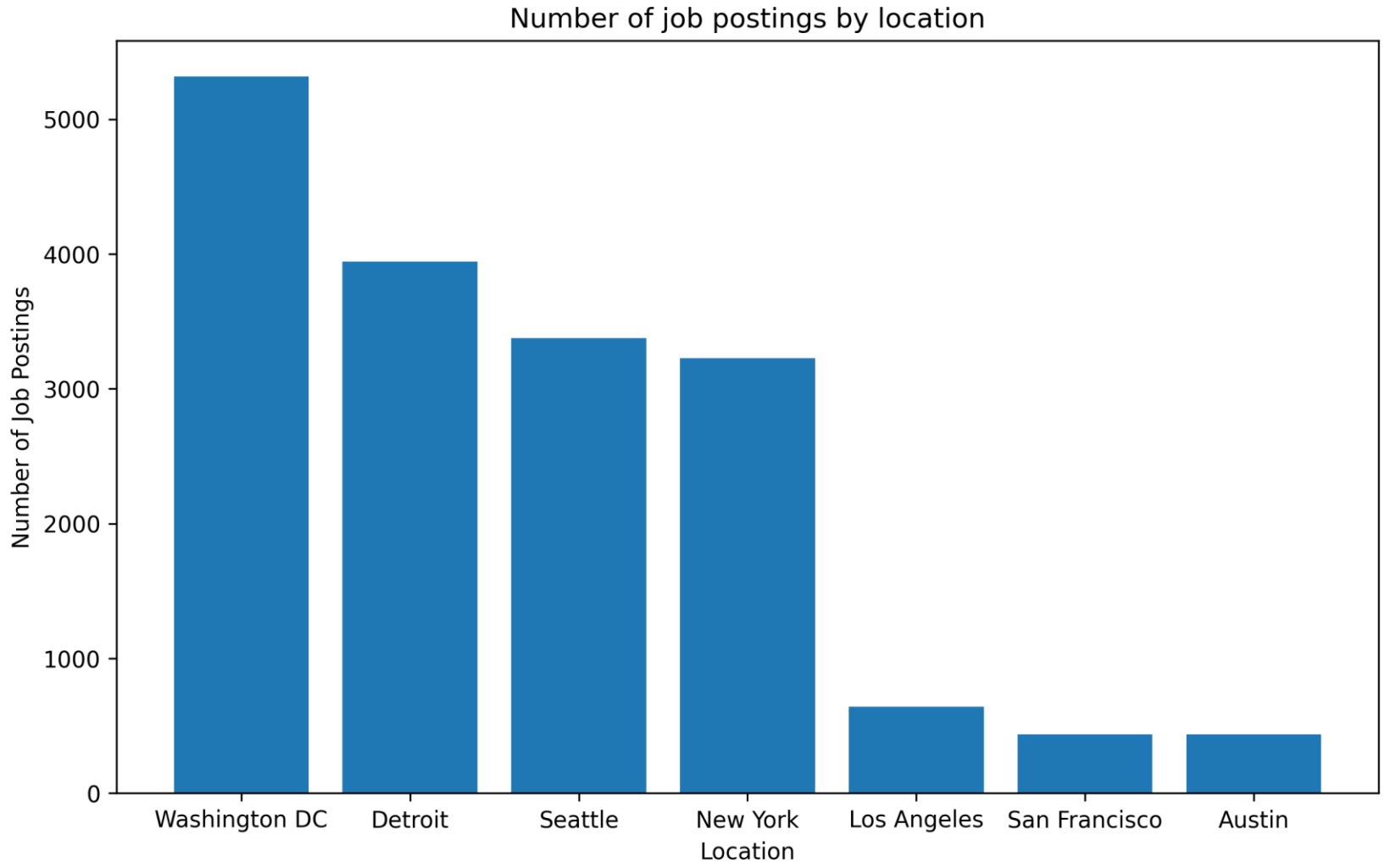
METHODOLOGY

- The data used in this report comes from:
  - Job postings dataset (see Appendix for more details)
  - Stack Overflow Developer Survey (see Appendix for more details)
- The Job postings dataset is accessed through a web API and did not need any cleaning & wrangling.
- The Stack Overflow Developer Survey (SODS) comes as a .csv Excel file and was cleaned of duplicated rows.
- The SODS cleaned file contains many columns with missing values, some with a great percentage of them.
- Dropping all SODS rows with at least one missing value would greatly reduce the size of the sample, so we did not do an overall data set wrangling to eliminate rows with missing values.
- We adopted the strategy of generating a specific, reduced SODS dataset for each analysis, with only the pertaining columns, and dropping rows with missing values in this reduced dataset.
- This approach allowed us to get the biggest sample size possible for each analysis without missing values.

Results

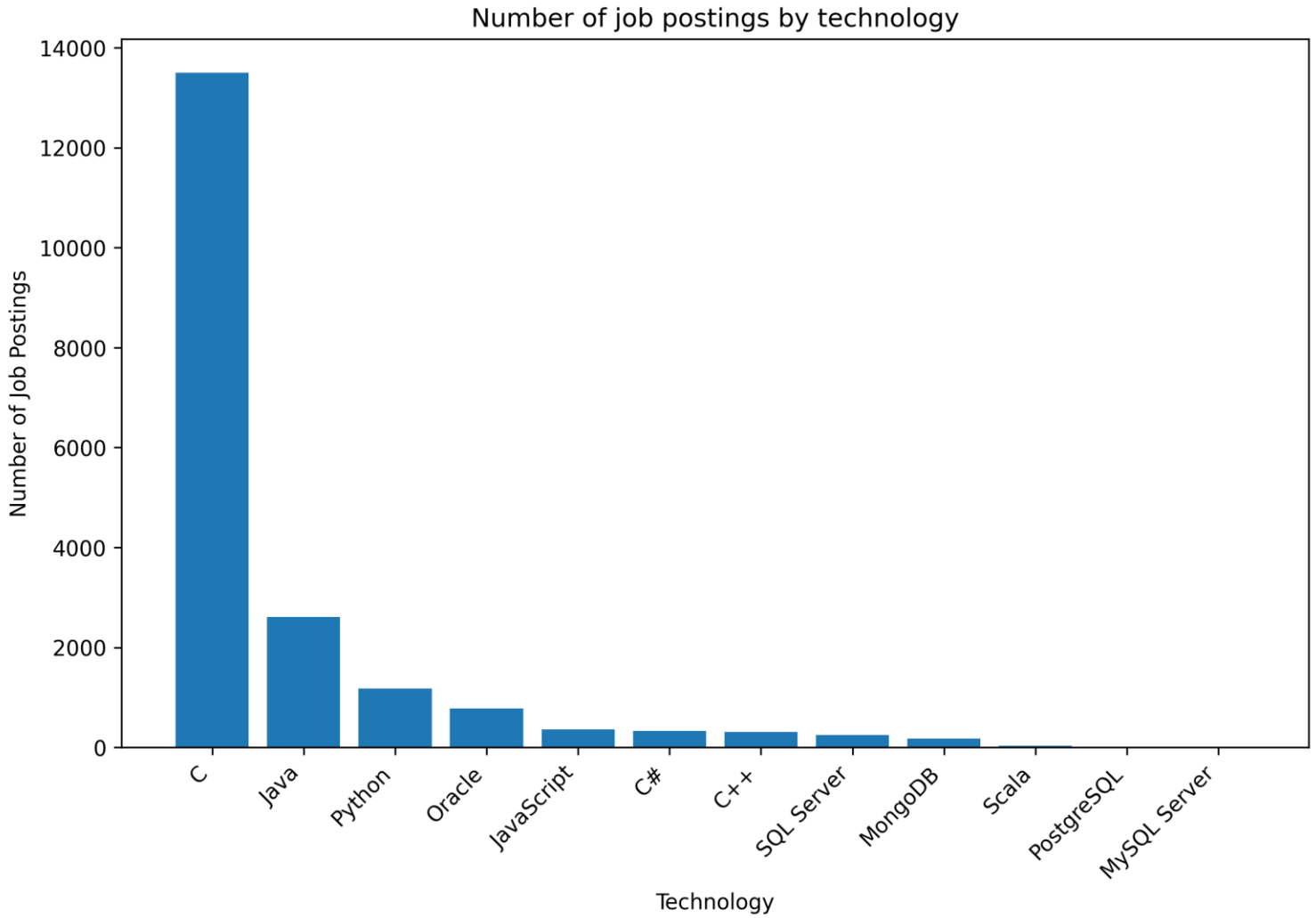


JOB POSTINGS BY LOCATION



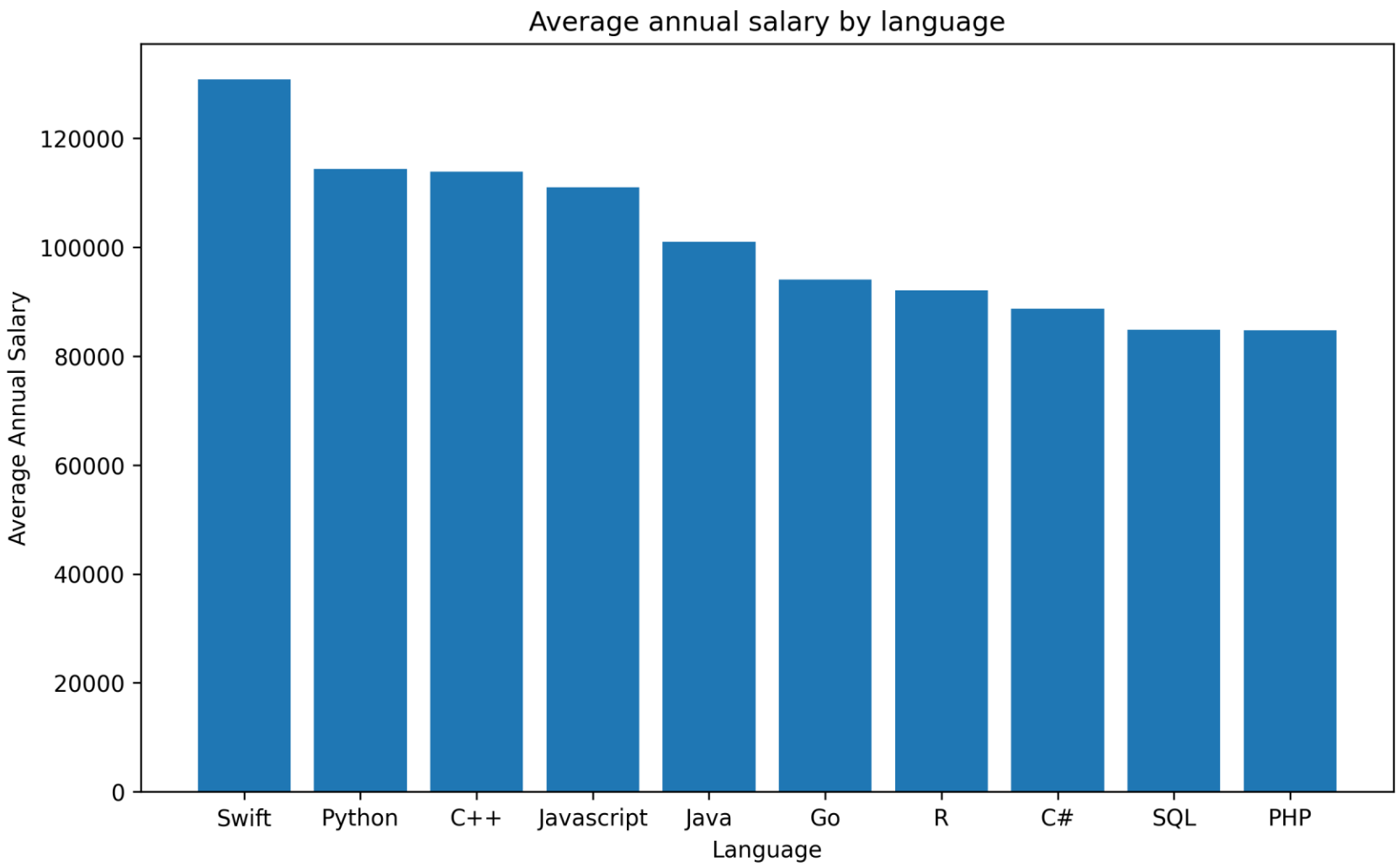
Washington DC is the location with the most job postings!

JOB POSTINGS BY TECHNOLOGY



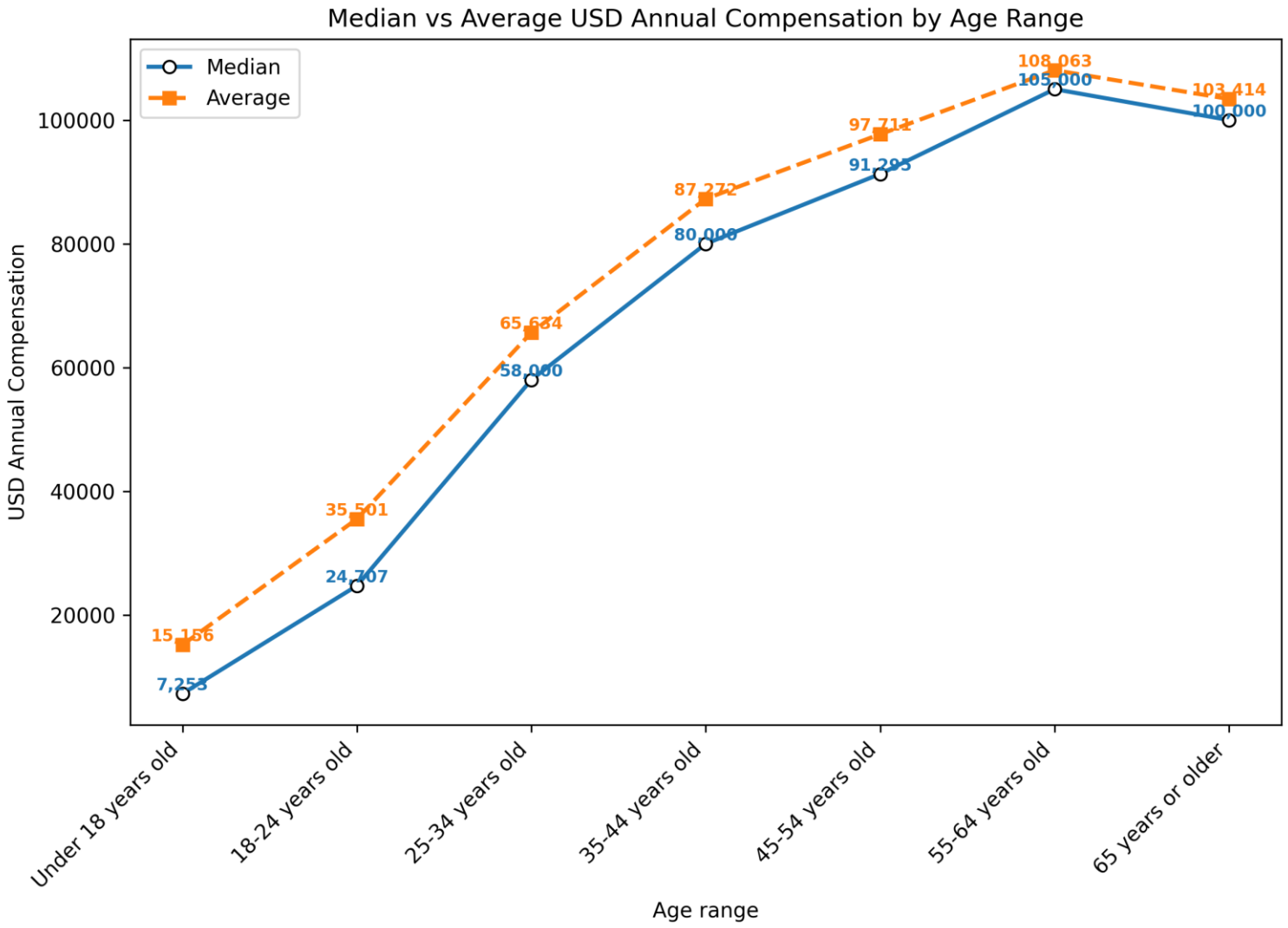
C is by far the location with the most job postings!

POPULAR LANGUAGES ANNUAL SALARY



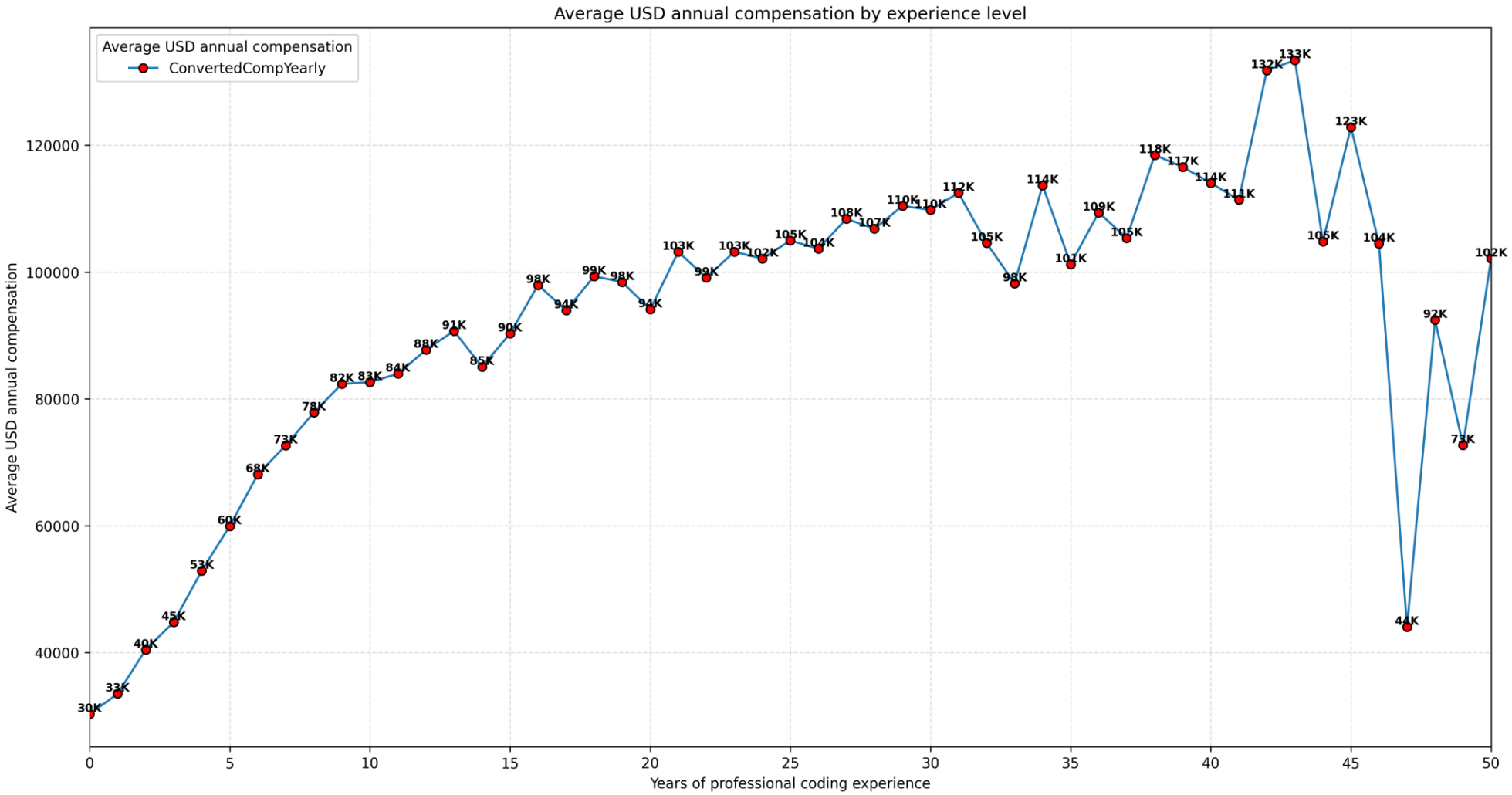
Swift and Python are the ones which yield better salaries, but the difference among languages is not too big!

Annual salary by age range



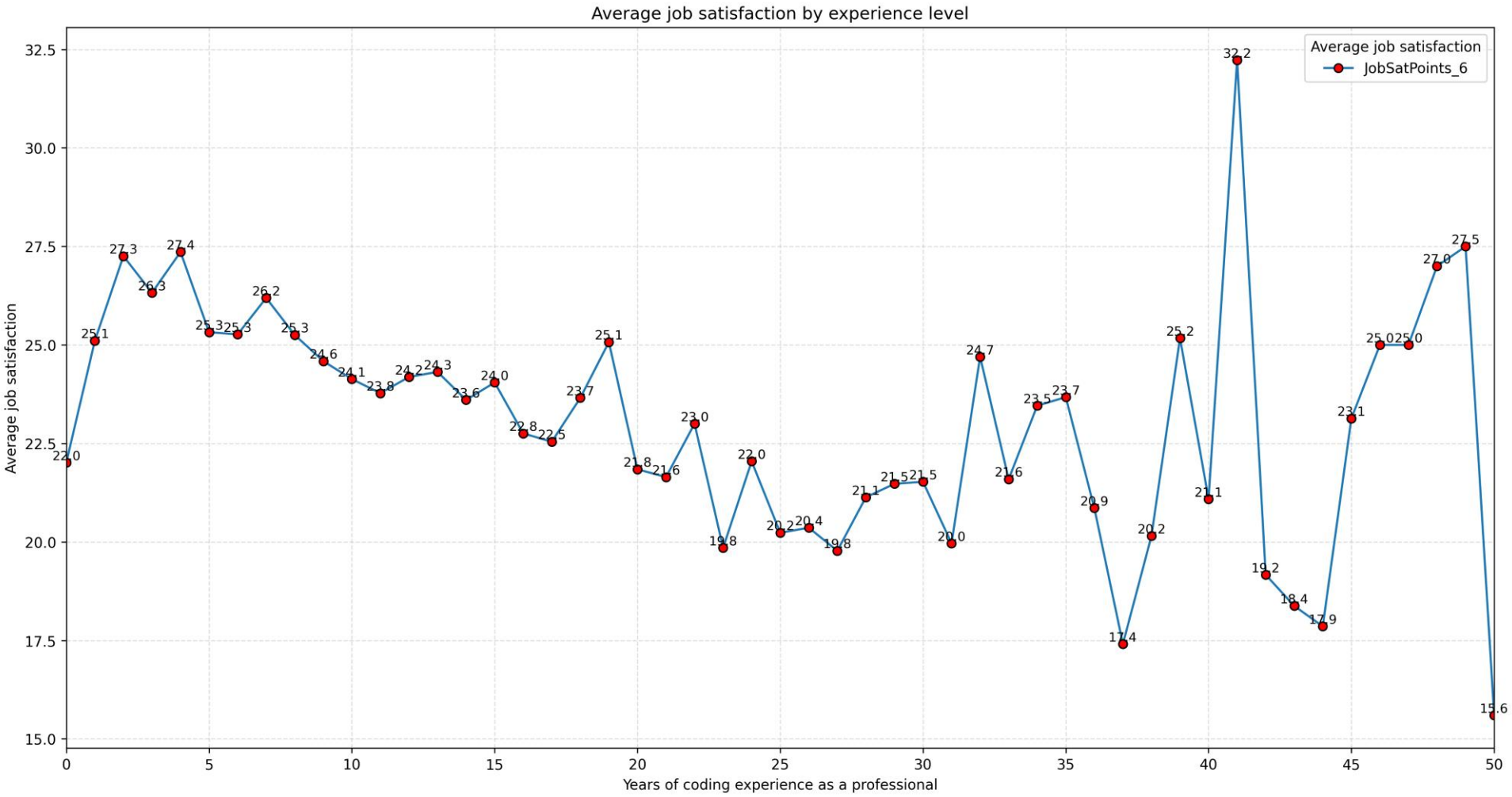
Annual salaries increase with age, which is expected!

Annual compensation by Experience level



Compensation increases with experience level, up to a point (44 years)!

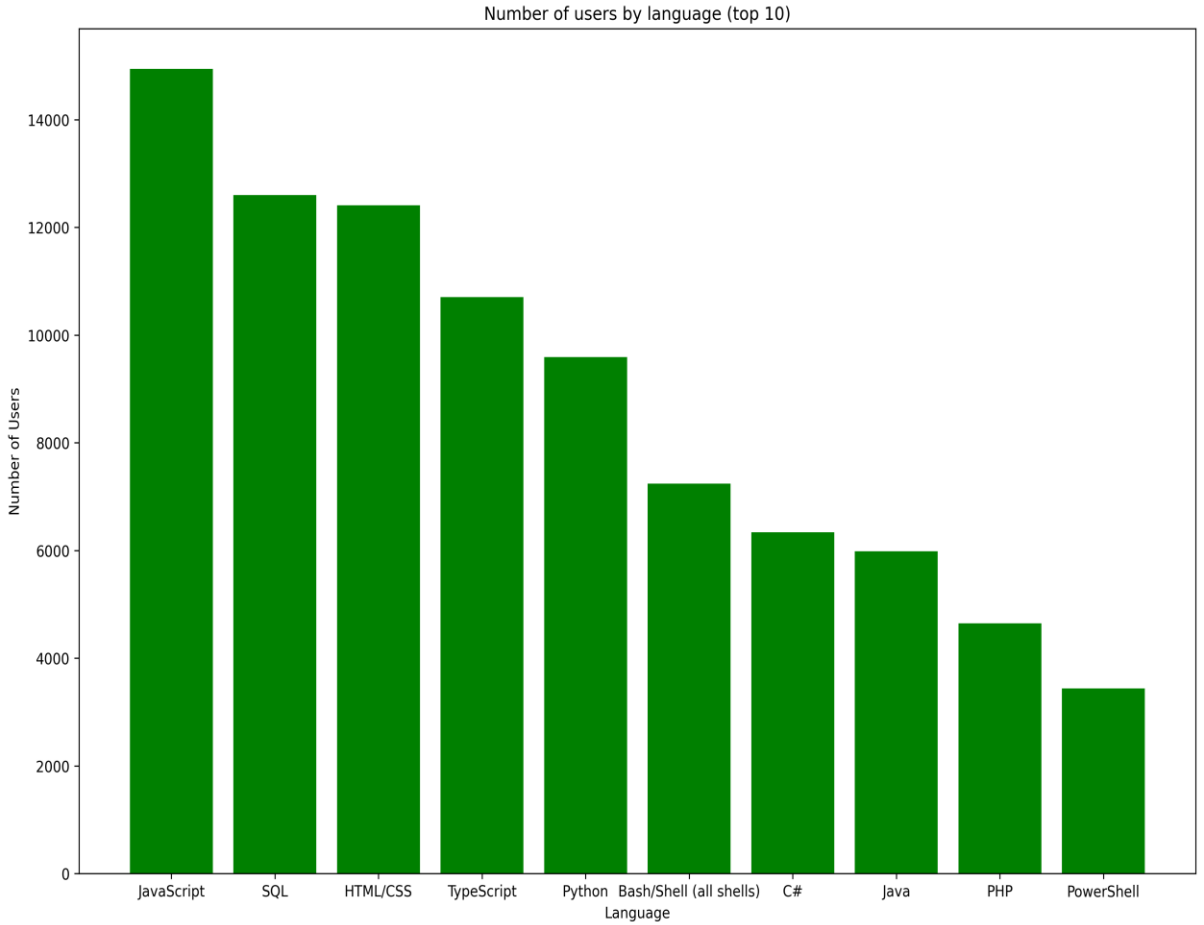
Job Satisfaction by Experience level



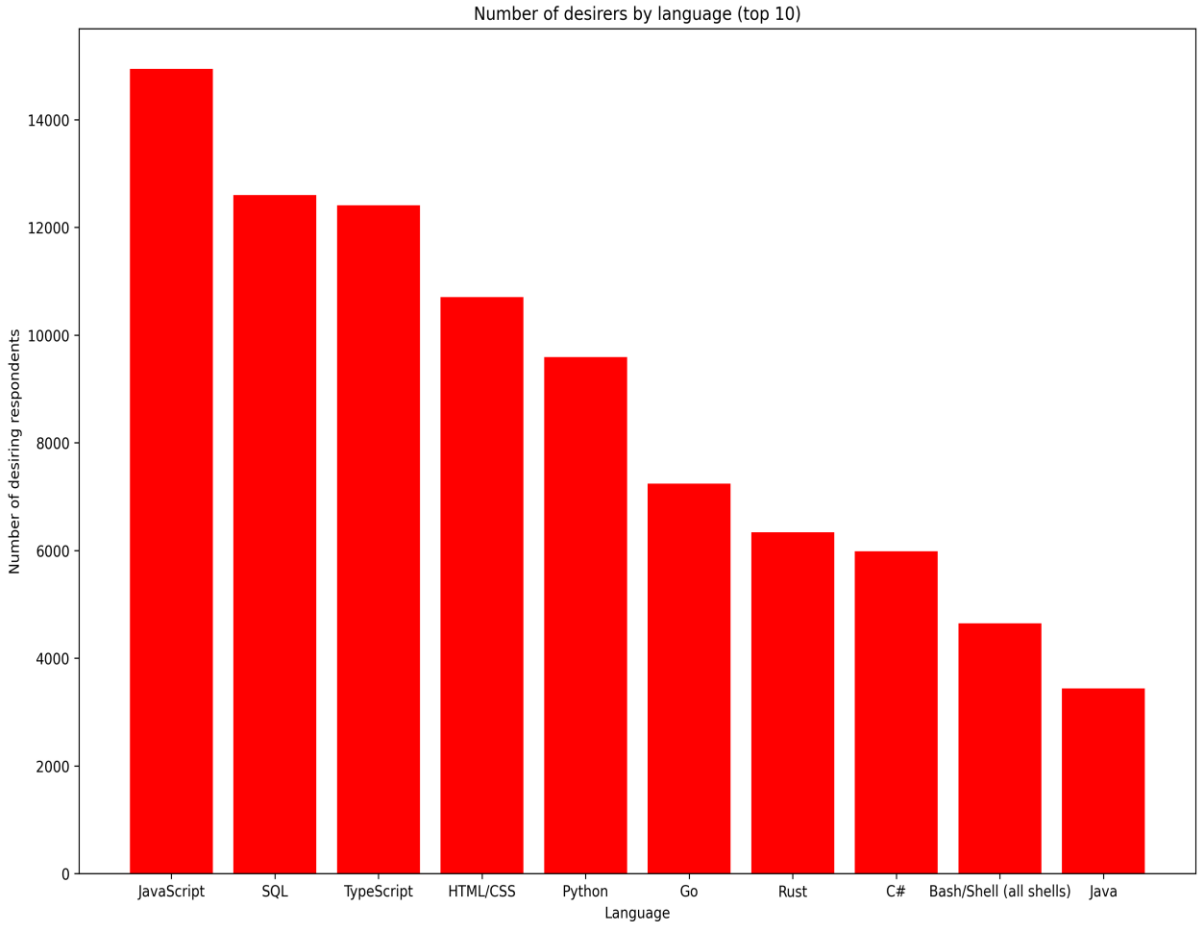
Job satisfaction seems quite low, in general!

PROGRAMMING LANGUAGE TRENDS

Current year : Number of users by language (top 10)



Next year : Number of desirers by language (top 10)



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

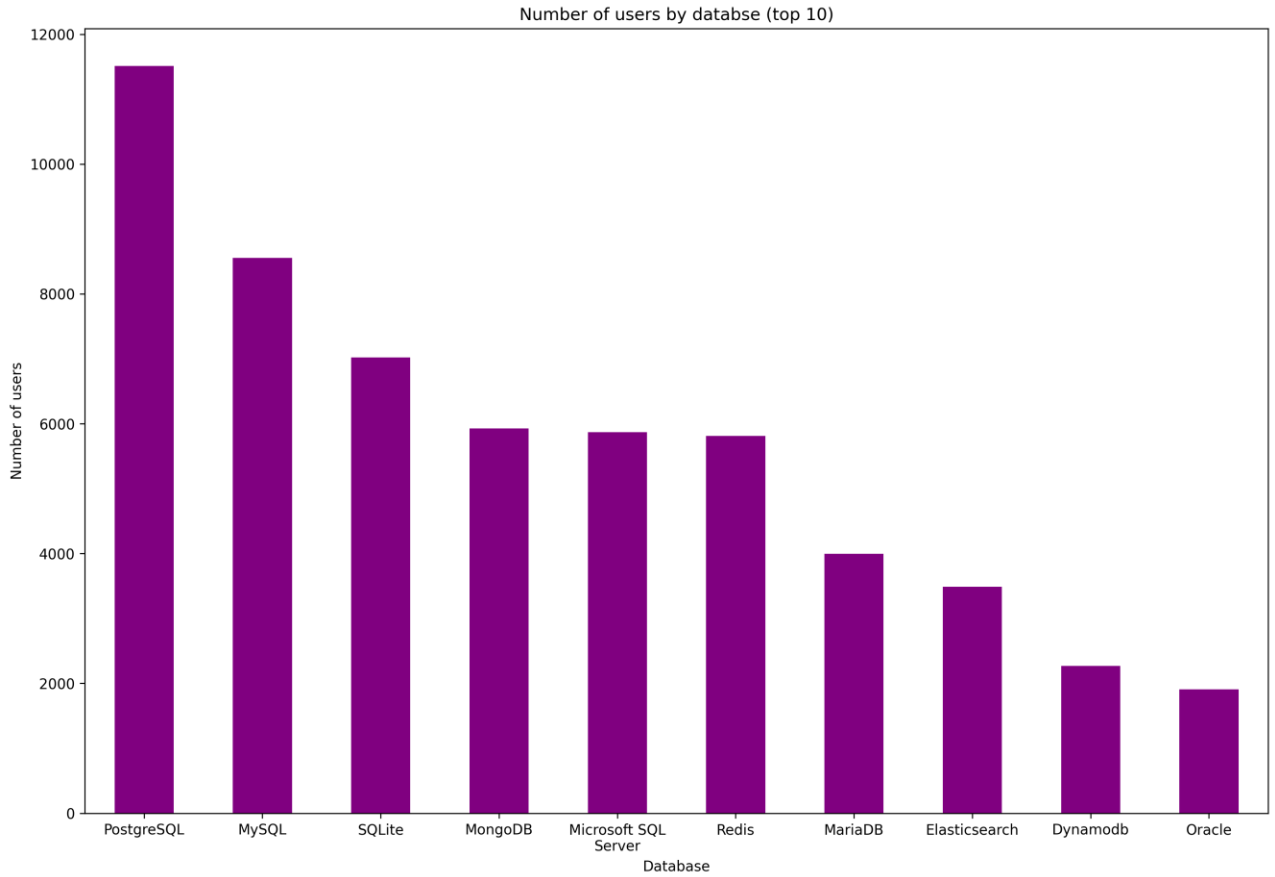
- JavaScript and SQL will continue as the most popular languages.
- PHP and PowerShell will be no longer in the top 10.
- For the other languages already in the top 10, future use will not have a significant change.
- Rust and Go will enter the top 10.

Implications

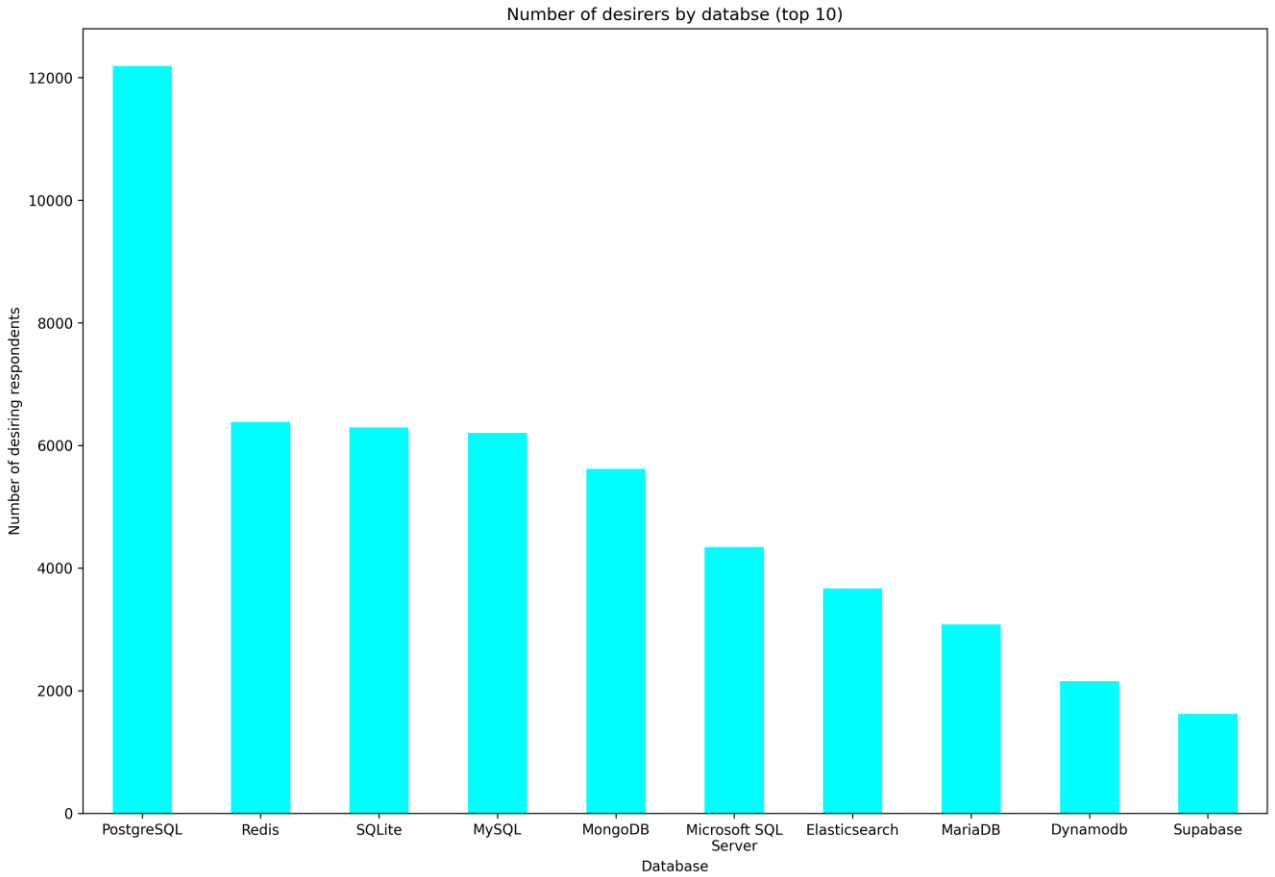
- Assess company's assets on Go and Rust, decide on possible recruiting action.
- Assess company's assets on PHP and PowerShell, decide on possible retraining action.

DATABASE TRENDS

Current year : Number of users by database (top 10)



Next year : Number of desirers by database (top 10)



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

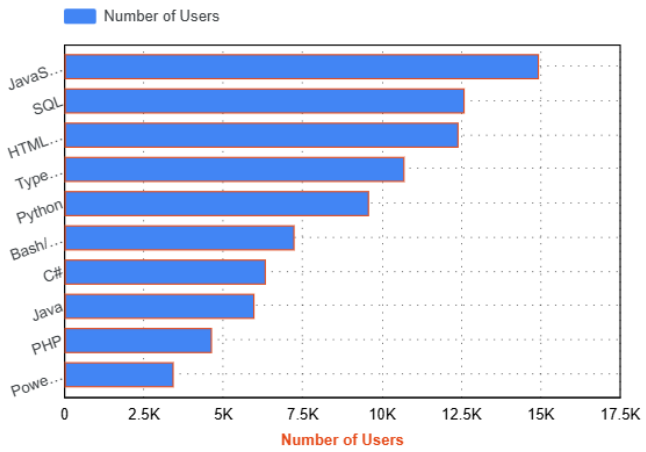
- PostgreSQL will continue to be the most popular database.
- MySQL will have a significant decrease in popularity.
- Oracle will leave the top 10.
- Supabase will enter the top 10.
- For the rest of the databases already in the top 10, future usage will not have a significant change.

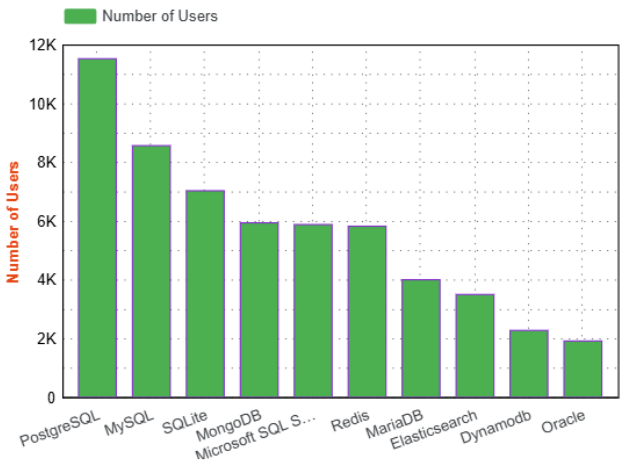
Implications

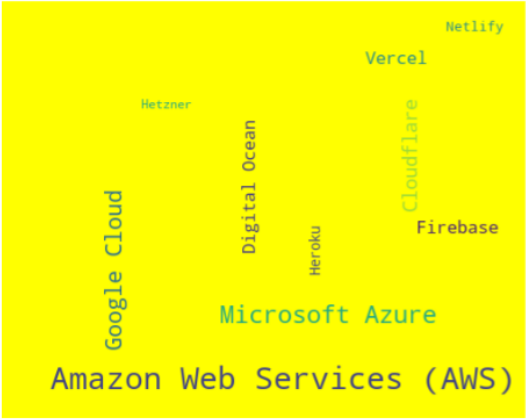
- Assess company's assets on Supabase, decide on possible recruiting action.
- Assess company's assets on MySQL and Oracle, decide on possible retraining action.

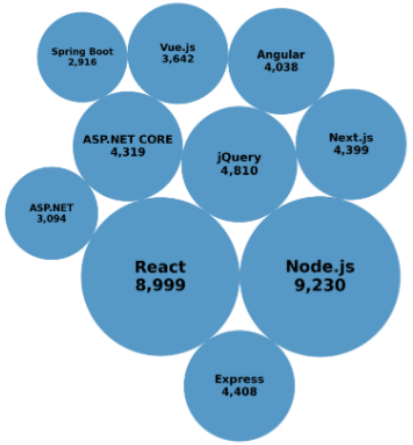
DASHBOARD

DASHBOARD TAB 1 : Current Technology Usage

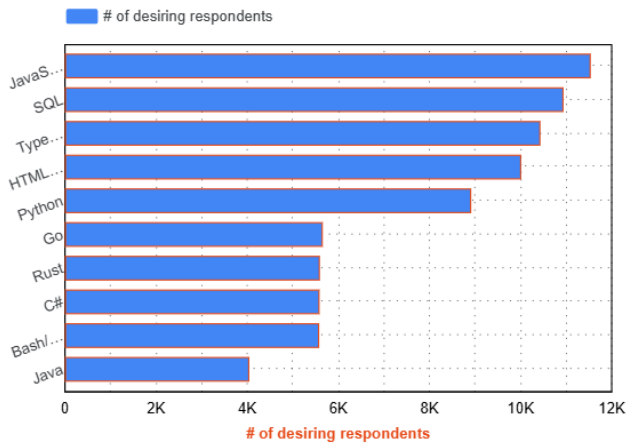


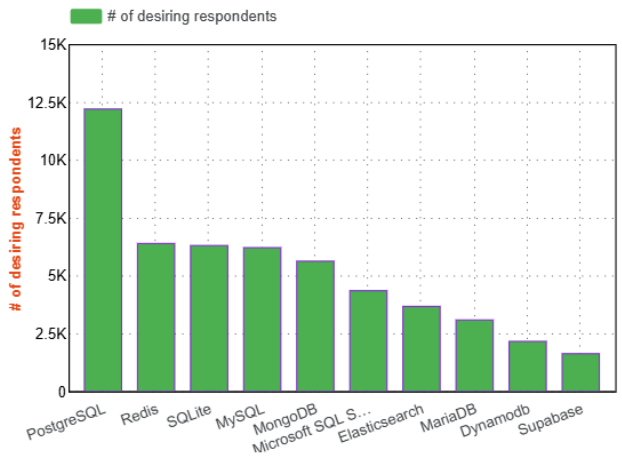


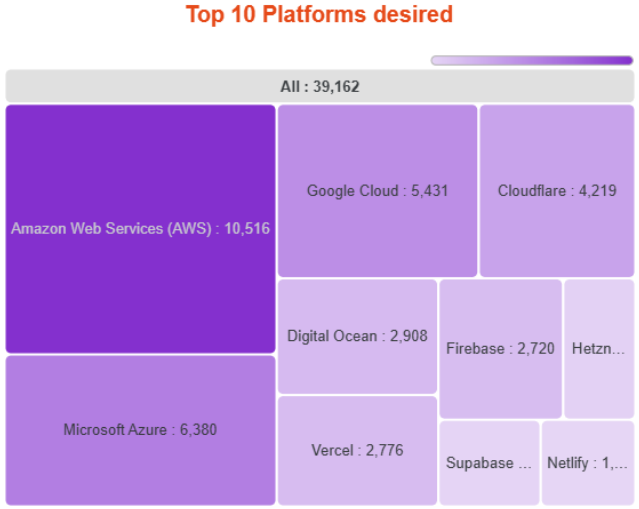


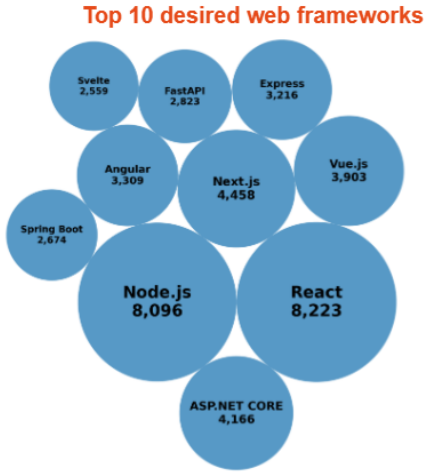


DASHBOARD TAB 2: Future Technology Trends

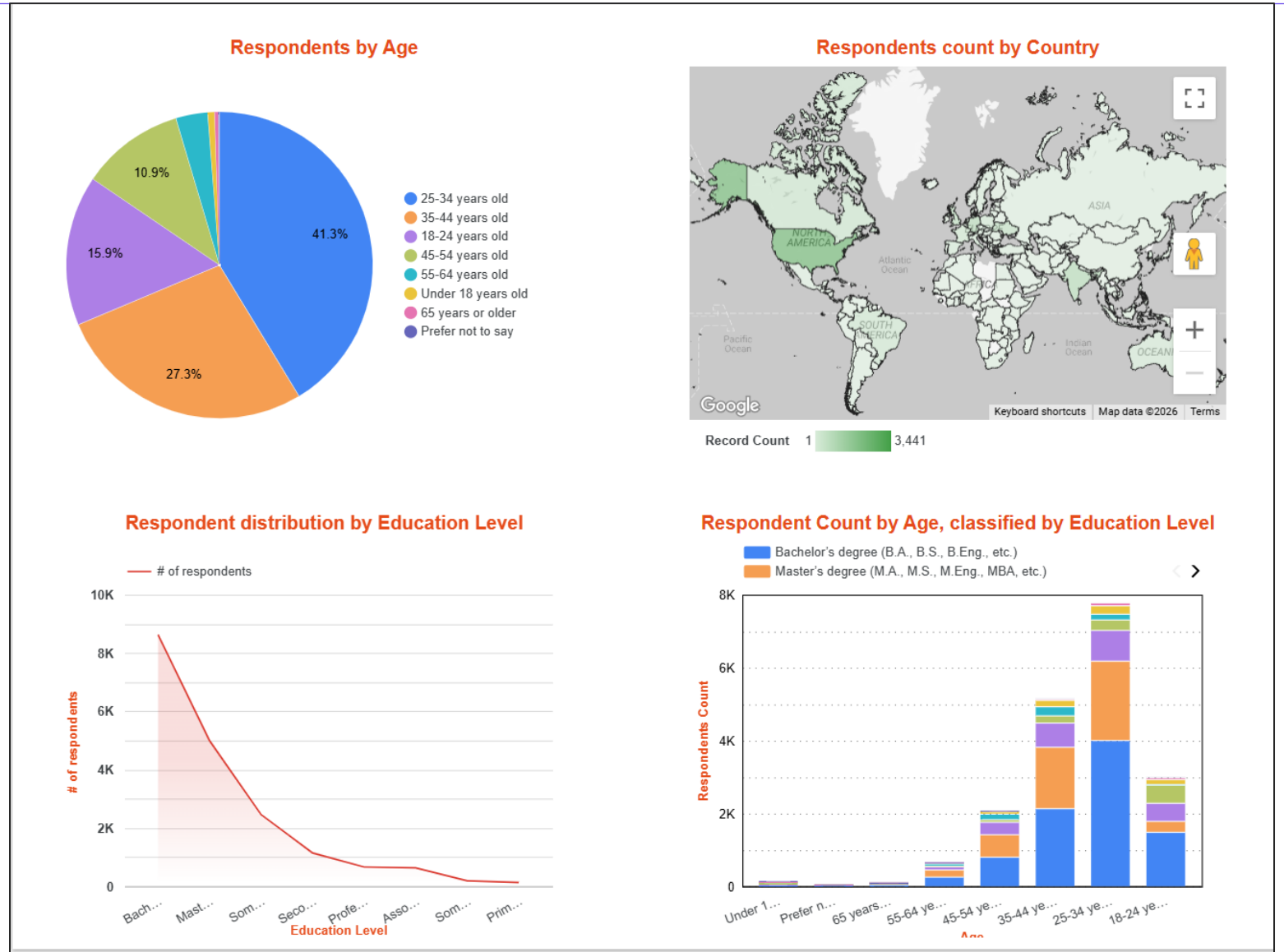








DASHBOARD TAB 3: Demographics



Skills Network

IBM

DISCUSSION: TOP 10 LANGUAGES

- JavaScript and SQL will continue as the most popular languages.
- PHP and PowerShell will be no longer in the top 10 languages. The company must assess its assets on them and decide on possible retraining actions.
- For the other languages already in the top 10, future usage will not present a significant change, so it seems there is no action to take regarding them.
- Rust and Go will enter the top 10 languages. The company must assess its assets on them and decide on possible recruitment actions.

DISCUSSION: TOP 10 DATABASES

- PostgreSQL will continue to be the most used database.
- MySQL will have a significant decrease on usage and Oracle will leave the top 10 databases. The company must assess its assets on them and decide on possible retraining action.
- Supabase will enter the top 10 databases. The company must assess its assets on it and decide on possible recruiting action.
- For the other databases already in the top 10, future usage will not present a significant change, so it seems there is no action to take regarding them.

DISCUSSION: TOP 10 CLOUD PLATFORMS

- AWS and Microsoft Azure will continue to be the most used platforms.
- Heroku will leave the top 10. The company must assess its assets on it and decide on possible retraining action.
- Supabase will enter the top 10 platforms. The company must assess its assets on it and decide on possible recruiting action.
- For the other platforms already in the top 10, future usage will not present a significant change, so it seems there is no action to take regarding them.

DISCUSSION: TOP 10 WEB FRAMEWORKS

- React and Node.js will continue to be the most used web frameworks but React will pass Node.js and be #1.
- jQuery and ASP.NET will leave the top 10. The company must assess its assets on it and decide on possible retraining action.
- FastAPI and Svelte will enter the top 10. The company must assess its assets on them and decide on possible recruiting action.
- For the other web frameworks already in the top 10, future usage will not present a significant change, so it seems there is no action to take regarding them.

DISCUSSION: DEMOGRAPHICS

- Almost 70% of developers have age between 25 and 44 years old and hold a Bachelor’s or Master’s degree. Company must assess how current work environment meets the expectations of this major group and decide on actions to change it, if needed.
- USA is by far the country with the biggest number of developers.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Usage of languages, databases, cloud platforms and web frameworks will change on the near future.
- Workforce demographics and education are quite concentrated on one specific profile.

Implications

- Workforce recruitment and retraining actions must be considered.
- Current work environment must be assessed to check on compatibility of major workforce demographic and education profile.

CONCLUSION: Workforce recruiting

- Company must assess its assets and decide on recruiting actions for the following technologies:
  - Result
  - Go
  - Supabase
  - FastAPI
  - Svelte

CONCLUSION: Workforce retraining

- Company must assess its assets and decide on retraining actions for the following technologies:
  - PHP
  - PowerShell
  - MySQL
  - Oracle
  - Heroku
  - Jquery
  - ASP.NET

CONCLUSION: Work environment

- Almost 70% of developers have age between 25 and 44 years and hold a Bachelor’s or Master’s degree. Company must assess how current work environment meets the expectations of this major group and decide on actions to change it, if needed.

APPENDIX

- Datasets used in this report:
  - Job Postings https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/module 1/Accessing Data Using APIs/jobs.json
  - Stack Overflow Developer Survey: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/HLOosvsPgIwt5dgOOh1RSg/survey-data-updated.csv

    https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/n01PQ9pSmiRX6520flujwQ/survey-data.csv